# Countable Additivity is Not First-Order Axiomatizable

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Abstract. In the first-order language of Boolean algebras equipped with a normalized finitely additive charge, no first-order theory characterizes those models whose charge extends to a sigma-additive measure on the generated sigma-algebra. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras. The proof uses a single ultraproduct argument via Łoś's theorem: Dirac charges on the finite-cofinite algebra are individually sigma-additive, but their ultraproduct is the finite-cofinite charge, which fails sigma-additivity — contradicting Łoś's theorem if any first-order axiomatization existed.

A property of structures is *first-order axiomatizable* in a language $L$ if there exists a set of $L$-sentences whose models are exactly the structures satisfying it. Łoś's theorem gives the key diagnostic: if a property fails in an ultraproduct of structures that individually satisfy it, no first-order axiomatization exists.

Countable additivity distinguishes $\sigma$-additive measures from finitely additive charges. That it cannot be expressed by a single first-order sentence is clear: the condition quantifies over countable sequences, unavailable in first-order logic. Its non-first-order character is treated as background knowledge in the probability logic literature [**3, 4, 2**], but the sharper question — whether countable additivity admits a first-order *axiomatization* by any set of sentences, not merely a single formula — does not appear to have been settled by an explicit proof.

We settle it negatively.

Let $\mathcal{L}_{\mathrm{BA},\mu}$ be the one-sorted first-order language with:

- Boolean operations $\wedge, \vee, \neg, 0, 1$;
- for each rational $q \in [0,1] \cap \mathbb{Q}$, unary relation symbols $M_{\leq q}(x)$ and $M_{\geq q}(x)$, intended to mean $\mu(x) \leq q$ and $\mu(x) \geq q$ respectively.

A structure for $\mathcal{L}_{\mathrm{BA},\mu}$ is a Boolean algebra $B$ together with a $[0,1]$-valued finitely additive probability $\mu$, encoded through the truth of the predicates $M_{\leq q}$ and $M_{\geq q}$. Using rational comparison predicates rather than a function symbol into $[0,1]$ avoids introducing a second numeric sort.

Definition 1 (First-order theory of finitely additive probabilities). Let $T_{\mathrm{fa}}$ be the $\mathcal{L}_{\mathrm{BA},\mu}$-theory axiomatizing:

(i) $B$ is a Boolean algebra;

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(ii) normalization: $\mu(0) = 0$ and $\mu(1) = 1$;
(iii) monotonicity: $x \leq y \Rightarrow \mu(x) \leq \mu(y)$;
(iv) finite additivity: for all rationals $r, s$ with $r + s \leq 1$,

$$x \wedge y = 0 \wedge M_{\leq r}(x) \wedge M_{\leq s}(y) \ \Rightarrow\ M_{\leq r+s}(x \vee y),$$

and similarly for lower bounds.

The class of Boolean algebras with normalized finitely additive probability is first-order axiomatizable in $\mathcal{L}_{\mathrm{BA},\mu}$. Countable additivity, by contrast, requires quantification over an arbitrary countable sequence $(a_n)_{n\in\mathbb{N}}$ — not available in first-order logic — and so is not a sentence of $\mathcal{L}_{\mathrm{BA},\mu}$.

Proposition 2 ($\sigma$-additive extension is not first-order axiomatizable). *There is no first-order $\mathcal{L}_{\mathrm{BA},\mu}$-theory $T \supseteq T_{\mathrm{fa}}$ such that, for every $(B, \mu) \models T_{\mathrm{fa}}$,*

$$(B, \mu) \models T \quad \Longleftrightarrow \quad \mu \text{ extends to a } \sigma\text{-additive measure on } \sigma(B).$$

*Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras.*

Proof. Let $E$ be the Boolean algebra of finite-cofinite subsets of $\mathbb{Q}$, fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$, and let $\delta_n$ be the Dirac charge at $q(n)$. Each $(E, \delta_n)$ is a model of $T_{\mathrm{fa}}$ in which $\delta_n$ is $\sigma$-additive. Suppose for contradiction that a theory $T$ as described exists; then each $(E, \delta_n) \models T$.

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter. By Łoś's theorem [**5**], the ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ also models $T$. The charge induced on the diagonal copy of $E$ is

$$\ell(A) = \lim_{\mathcal{U}} \delta_n(A) = \lim_{\mathcal{U}} \mathbf{1}_{q(n)\in A}.$$

If $A$ is finite, $q(n) \in A$ for only finitely many $n$, so $\ell(A) = 0$. If $A$ is cofinite, $q(n) \in A$ for cofinitely many $n$, so $\ell(A) = 1$ since $\mathcal{U}$ extends the cofinite filter. Thus $\ell$ is the finite-cofinite charge.

The sequence $A_k = \mathbb{Q}\setminus\{q(0), \ldots, q(k)\}$ satisfies $A_0 \supseteq A_1 \supseteq \cdots$ and $\bigcap_k A_k = \varnothing$, yet $\ell(A_k) = 1$ for all $k$. Hence the charge on the diagonal copy of $E$ fails $\sigma$-additivity and does not extend to a $\sigma$-additive measure on $\sigma(E)$. Consequently the ultraproduct cannot satisfy any theory $T$ characterizing $\sigma$-additive extension, contradicting Łoś's theorem applied to the assumption that each $(E, \delta_n) \models T$. $\square$

Remark 3. The ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ is an $\mathcal{L}_{\mathrm{BA},\mu}$-structure in its own right; $\ell$ is the charge induced on the diagonal copy of $E$ inside it. Identifying the ultraproduct with $(E, \ell)$ is harmless for the contradiction, since the failure of $\sigma$-additivity is witnessed on the diagonal copy.

The proof produces one purely finitely additive charge as an ultraproduct of $\sigma$-additive ones; that single witness suffices for the non-axiomatizability conclusion. A natural follow-up question is which purely finitely additive charges on a Boolean algebra arise as pointwise ultralimits of $\sigma$-additive probabilities on the same algebra. On a $\sigma$-algebra the answer is none: any pointwise ultralimit of $\sigma$-additive probabilities is itself $\sigma$-additive, by Nikodym-type convergence [**1**], IV.9.8. On a general Boolean algebra the situation is more structured, and the problem is settled in several natural regimes. The remaining open case reduces to whether there exists a non-$\sigma$-complete non-atomic Boolean algebra admitting no $\sigma$-additive probability.

## References

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